

SUPPLY & DEMAND PROBLEM SET—SOLUTIONS

ECON101—INTRODUCTION TO MICROECONOMICS.

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Problem 1

To find the equilibrium in the chocolate market, we must first determine the consumer income Y from the labor market.

Step 1: Labor Market Equilibrium

Set $L_D = L_S$:

$$\begin{aligned}100 - 2w &= -20 + 4w \\120 &= 6w \implies w^* = 20\end{aligned}$$

Substitute w^* into L_D :

$$L^* = 100 - 2(20) = 60$$

Total Income: $Y = w^* \times L^* = 20 \times 60 = 1200$.

Step 2: Chocolate Market Equilibrium

Substitute $Y = 1200$, $P_{\text{caramel}} = 50$, $P_{\text{milk}} = 8$, and $P_{\text{cacao}} = 10$ into the equations:

$$\begin{aligned}Q_{\text{choc},D} &= 560 - 4P_{\text{choc}} + 0.1(1200) + 4(50) - 5(8) = 840 - 4P_{\text{choc}} \\Q_{\text{choc},S} &= -60 + 6P_{\text{choc}} - 10(10) = -160 + 6P_{\text{choc}}\end{aligned}$$

Set $Q_{\text{choc},D} = Q_{\text{choc},S}$:

$$\begin{aligned}840 - 4P_{\text{choc}} &= -160 + 6P_{\text{choc}} \\1000 &= 10P_{\text{choc}} \implies P_{\text{choc}}^* = 100 \\Q_{\text{choc}}^* &= 840 - 4(100) = 440\end{aligned}$$

Problem 2

(a) Initial Equilibria:

- Village 1: $200 - P = -40 + 5P \implies P_1^* = 40, Q_1^* = 160$.
- Village 2: $100 - 2P = 20 + 4P \implies P_2^* = 13.33, Q_2^* = 86.66$.

(b) and (c)—Arbitrage (30 units): Panqueque (see figure 1 below) buys in Village 2 (increasing demand) and sells in Village 1 (increasing supply).

- Village 2: $100 + 30 - 2P = 20 + 4P \implies P_2^*(30) = 18.33$.
- Village 1: $200 - P = -40 + 30 + 5P \implies P_1^*(30) = 35$.

(d) Profit: $(35 - 18.33) \times 30 = 500$.

Optimal Quantity: Now suppose that Panqueque carries the same arbitrage operation as above, but with x units. Panqueque buys in Village 2 x units (increasing demand) and sells in Village 1 x units (increasing supply).

- Village 2: $100 + x - 2P = 20 + 4P \implies P_2^*(x) = \frac{80+x}{6}$.
- Village 1: $200 - P = -40 + x + 5P \implies P_1^*(x) = \frac{240-x}{6}$.



Figure 1: Panqueque, the entrepreneur.

Therefore, the revenue function from Panqueque's enterprise, as a function of x is

$$\begin{aligned}
 R(x) &= (P_1^*(x) - P_2^*(x))x \\
 &= \left(\frac{240-x}{6} - \frac{80+x}{6} \right) x \\
 &= \left(\frac{160-2x}{6} \right) x \\
 &= \frac{80}{3}x - \frac{1}{3}x^2
 \end{aligned}$$

Maximizing $R(x)$ via the first-order condition:

$$R'(x) = \frac{80}{3} - \frac{2}{3}x = 0 \implies x^* = 40 \text{ units}$$

Max Profit: $R(40) \approx 533.33$.

Remark: There is no need to use calculus for this problem if you remember high-school analytic geometry. If a parabola has two different roots, its vertex occurs at the midpoint of the two roots. See Figure 2 below.

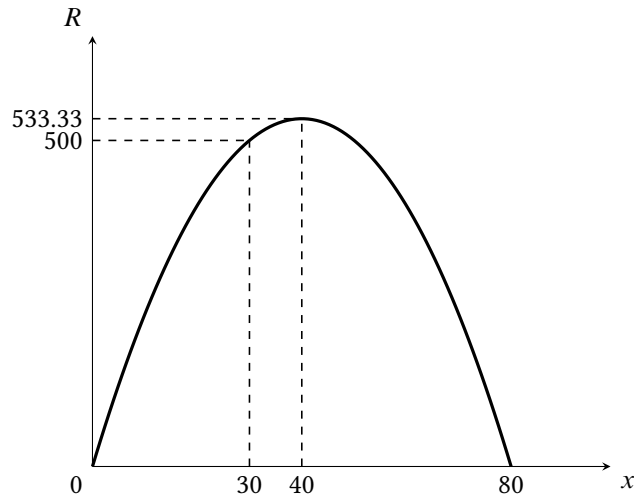
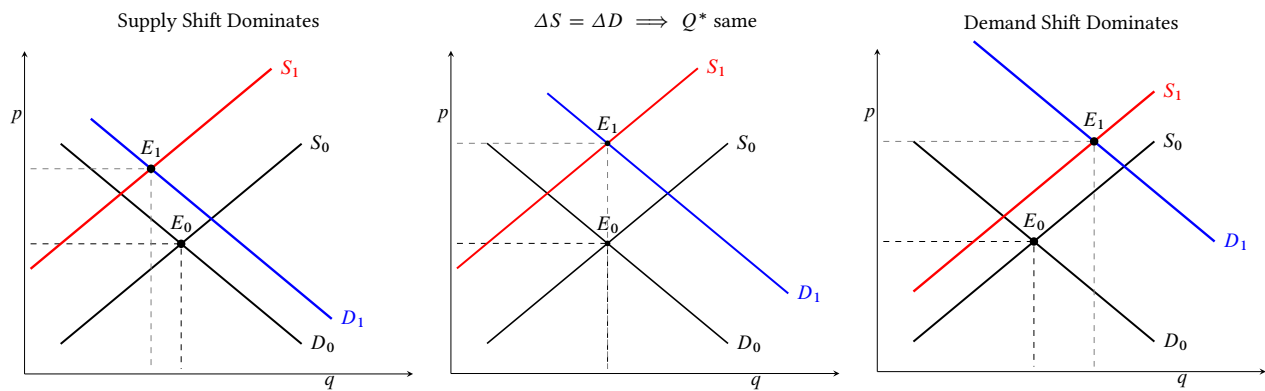


Figure 2: Diagram illustrating revenue as a function of x , $R(x)$, and its maximum.

Problem 3

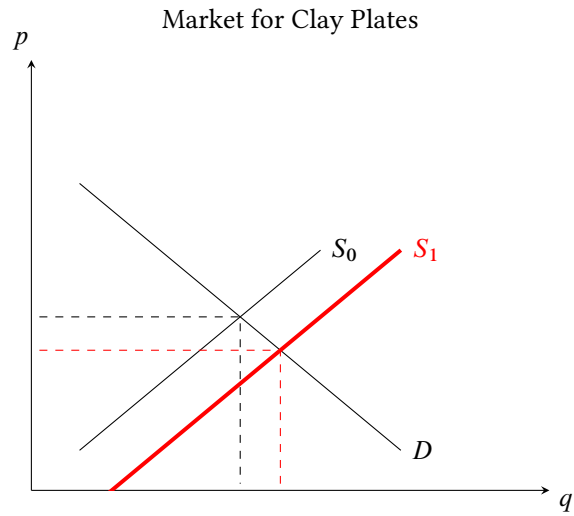
3a Kumis Market: Substitute Scare and Labor Cost Increase

A health report claiming Shubat is bad increases demand for Kumis ($D_0 \rightarrow D_1$). Simultaneously, higher wages for horse care workers decrease supply ($S_0 \rightarrow S_1$). **Result:** Equilibrium price (P^*) definitely rises; quantity (Q^*) is ambiguous.



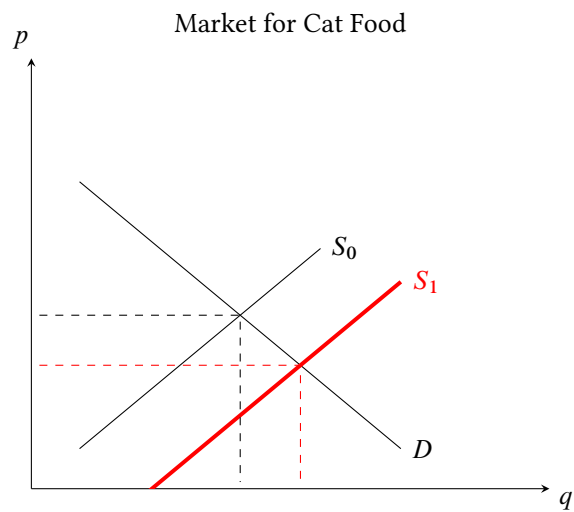
3b Clay Plates: Technological Advance

Cheaper clay shifts the supply curve to the right ($S_0 \rightarrow S_1$). **Result:** P^* falls, Q^* rises.



3c Cat Food: Market Entry

Two new firms entering the market shift the supply curve to the right ($S_0 \rightarrow S_1$). **Result:** P^* falls, Q^* rises.



3d Bus Rides: Preference Shift to Walking

The health report shifts preferences away from buses, decreasing demand ($D_0 \rightarrow D_1$). **Result:** Both P^* and Q^* fall.

